SOC ANALYST PRACTICAL LABS

Topic 1 — **Advanced Log Analysis**

Topic 2 — **Threat Intelligence Integration**

Topic 3 — **Incident Escalation Workflows**

TOPIC 1

Advanced Log Analysis

Overview

Log analysis is the cornerstone of Security Operations Center (SOC) work. Security logs are generated by every device on a network — firewalls, endpoints, servers, applications, and cloud services. A skilled analyst must be able to collect, normalize, correlate, and interpret these logs to detect threats, investigate incidents, and produce actionable intelligence. This topic covers three core areas: log correlation, anomaly detection, and log enrichment.

1.1 Log Correlation

Log correlation is the process of combining events from multiple sources to identify patterns that indicate an attack or policy violation that would not be visible by looking at any single log stream.

Core Concepts

- Multi-source correlation: Combining firewall, endpoint, application, and authentication logs
- Time-based correlation: Events that occur in a specific sequence or within a time window
- Entity-based correlation: Grouping events by shared attributes — same IP, user, hostname
- Attack chain reconstruction: Mapping correlated events to MITRE ATT&CK; techniques

Classic Correlation Example: Brute Force + Exfiltration

Step	Event Source	Event Type	Indicator
1	Windows Security Log	Event ID 4625 — Failed Logon	Multiple failures from same IP
2	Windows Security Log	Event ID 4624 — Successful Logon	Success after failures = brute force
3	Firewall Log	Outbound TCP Port 53	DNS query immediately after login
4	DLP / Proxy Log	Large outbound transfer	Data staged for exfiltration
■ **Rule Example**	Correlation Rule: IF Event ID 4625 occurs >5 times from same source.ip within 60 seconds AND THEN Event ID 4624 occurs from same source.ip — THEN classify as Brute Force Success (T1110).		

Key Elastic KQL Queries for Lab

Correlate failed logins with outbound DNS:

```
event.code:4625 AND destination.port:53
```

Find all failed logins from a specific IP:

```
event.code:4625 AND source.ip:192.168.1.100
```

Show high-volume outbound events:

```
network.bytes > 1000000 AND network.direction:outbound
```

References & Learning Resources

- SANS Reading Room: 'Effective Log Management' — https://www.sans.org/reading-room/
- Elastic Documentation: Kibana KQL Reference — https://www.elastic.co/guide/en/kibana
- CISA Alert on Log Analysis: https://www.cisa.gov/resources-tools/resources

1.2 Anomaly Detection

Anomaly detection identifies events that deviate significantly from a baseline of normal behavior. Unlike signature-based detection (which looks for known bad patterns), anomaly detection can find novel attacks by flagging statistically unusual activity.

Types of Anomaly Detection

Type	Method	Example	Best For
Statistical	Threshold / Z-score	bytes_out > 2 standard deviations	Volume-based attacks
Rule-Based	IF-THEN logic	Login at 3AM from new country	Policy violations
ML-Based	Unsupervised clustering	Elastic ML Jobs on login times	Zero-day behavior
Behavioral	User Entity Profiling (UEBA)	Admin accessing HR files	Insider threats

Elastic Threshold Rule Configuration

Navigate to: Security > Rules > Create Rule > Threshold

```
Query: network.bytes > 1048576 Group by: source.ip Threshold: 1 occurrence Schedule:
Every 1 minute | Lookback: 1 minute
```

1.3 Log Enrichment

Raw logs often contain minimal context — an IP address with no location, a user account with no role information. Log enrichment adds contextual data to make logs more actionable for analysts.

Enrichment Type	What It Adds	Tool/Source	Example
GeoIP	Country, city, latitude/longitude	MaxMind GeoLite2 (built into Elastic)	8.8.8.8 → Mountain View, US
Threat Intel	Known-bad classification	AlienVault OTX, VirusTotal	185.220.101.1 → Tor Exit Node
Enrichment Type	**What It Adds**	**Tool/Source**	**Example**
User Context	Department, role, manager	LDAP / Active Directory	jsmith → Finance Dept, Manager: ajones
Asset Context	Owner, OS, criticality	CMDB / Asset database	192.168.1.50 → Finance Server, Critical
DNS Enrichment	Hostname from IP	Reverse DNS lookup	8.8.8.8 → dns.google

GeoIP Ingest Pipeline JSON (Elastic)

```
{ "processors": [ { "geoip": { "field": "source.address", "target_field": "source.geo",
"database_file": "GeoLite2-City.mmdb" } } ] }
```

■ Why Enrichment Matters	Key insight: A single log showing 'failed login from 8.8.8.8' is low-value. The same log enriched with GeoIP (Mountain View, US, Google LLC) + threat intel (benign DNS service) + user context (admin account, 3AM login time) becomes a high-fidelity detection with clear context.

TOPIC 2

Threat Intelligence Integration

Overview

Threat intelligence (TI) transforms raw security data into actionable knowledge about threats. A SOC analyst uses threat intelligence to contextualize alerts, validate IOCs, prioritize incidents, and hunt proactively for adversary activity. This topic covers TI types, the STIX/TAXII standard, practical integration with SIEMs, and MITRE ATT&CK-based; threat hunting.

2.1 Types of Threat Intelligence

Type	Focus	Time Horizon	Example Use
Strategic	Business risk, threat landscape	Months-years	Board reporting on APT groups targeting your industry
Operational	Ongoing campaigns, TTPs	Weeks-months	APT29 is actively phishing finance sector with CVE-2024-XXXX
Tactical	TTPs and adversary behavior	Days-weeks	Using MITRE T1078 to hunt for valid account abuse
Technical	IOCs: IPs, hashes, domains	Hours-days	Block 185.220.101.1 — known Tor exit used for C2

2.2 Indicators of Compromise (IOCs)

IOCs are forensic artifacts that indicate a system has been compromised or targeted. They are the most common form of technical threat intelligence and form the basis of automated detection and blocking.

IOC Type	Description	Example	Where to Check
IP Address	Known malicious source/destination	185.220.101.45 (Tor exit)	OTX, VirusTotal, Shodan
Domain / URL	C2 domains, phishing sites	malware-update.com	OTX, URLScan.io, VirusTotal

File Hash (MD5/SHA256)	Malware fingerprint	3a4b5c... (Cobalt Strike)	VirusTotal, MalwareBazaar
Email Address	Phishing sender	ceo@fake-company.com	EmailRep.io, Have I Been Pwned
Registry Key	Persistence mechanism	HKLM\Run\malware.exe	Wazuh rules, Sysmon events
User-Agent String	Malware C2 identification	Mozilla/5.0 (C2Bot)	Proxy/web gateway logs

2.3 STIX and TAXII Standards

STIX (Structured Threat Information eXpression) is a standardized language for representing cyber threat intelligence. TAXII (Trusted Automated eXchange of Intelligence Information) is the transport protocol for sharing STIX data between organizations and tools.

Standard	Purpose	Version	Key Use Case
STIX 2.1	Defines intelligence objects and relationships	Current	Representing APT campaigns, TTPs, IOCs in JSON
TAXII 2.1	Protocol for distributing STIX over HTTPS	Current	Automated feed ingestion into SIEMs
MISP	Open-source platform for sharing indicators	Active	Community-driven IOC sharing
OpenCTI	Open-source threat intel platform	Active	Aggregating and contextualizing TI feeds

2.4 AlienVault OTX — Key API Endpoints

Fetch malicious IP indicators:

```
curl -H 'X-OTX-API-KEY: YOUR_KEY' \
https://otx.alienvault.com/api/v1/indicators/export # Check specific IP reputation:
curl -H 'X-OTX-API-KEY: YOUR_KEY' \
https://otx.alienvault.com/api/v1/indicators/IPv4/185.220.101.45/general
```

2.5 MITRE ATT&CK; Framework

The MITRE ATT&CK; framework is a globally accessible knowledge base of adversary tactics and techniques based on real-world observations. It provides a common vocabulary for SOC teams to describe, detect, and respond to threat actor behavior.

Tactic (Goal)	ID	Example Technique	Detection Approach
Initial Access	TA0001	T1078 – Valid Accounts	Monitor logins outside business hours
Execution	TA0002	T1059 – Command & Scripting	Alert on encoded PowerShell, wscript
Persistence	TA0003	T1053 – Scheduled Task	Monitor Task Scheduler event logs
Privilege Escalation	TA0004	T1068 – Exploit Vulnerability	Alert on process elevation events
Tactic (Goal)	**ID**	**Example Technique**	**Detection Approach**
Defense Evasion	TA0005	T1070 – Log Clearing	Alert on Event ID 1102 (audit log cleared)
Discovery	TA0007	T1082 – System Info Discovery	Monitor systeminfo, hostname commands
Lateral Movement	TA0008	T1021 – Remote Services	Monitor RDP, SMB, WinRM connections
Exfiltration	TA0010	T1048 – Exfil over C2	Monitor large outbound transfers

T1078 Threat Hunt Query (Wazuh / Elastic)

```
# Hunt for logins by non-system accounts: user.name:* AND NOT user.name:system # Hunt for logins from outside the corporate network: event.action:logged-in AND NOT source.ip:192.168.0.0/16 # Wazuh MITRE-tagged alerts: rule.mitre.technique:T1078
```

■ **Key Distinction**	Threat Hunting vs Alert Triage: Alert triage is reactive (you respond to what fired). Threat hunting is proactive (you search for adversary behavior that did NOT trigger an alert). The best SOC teams do both. Use ATT&CK; techniques as your hunting hypotheses.

TOPIC 3

Incident Escalation Workflows

Overview

Incident escalation is the structured process of moving an incident from initial detection through triage, containment, investigation, and resolution. Effective escalation ensures the right people are involved at the right time, communication is clear and documented, and response actions are coordinated across the organization.

3.1 SOC Tier Structure

Tier	Role	Responsibility	Escalation Trigger
Tier 1	SOC Analyst (L1)	Monitor alerts, initial triage, classify severity, document basics	Confirmed threat, needs deep investigation, or exceeds Tier 1 skill
Tier 2	SOC Analyst (L2) / IR Specialist	Deep forensic analysis, malware reversing, IOC hunting, containment	Active compromise, multiple systems, data loss, or needs management briefing
Tier 3	Threat Hunter / IR Lead	Advanced threat hunting, zero-day analysis, APT investigations	Nation-state activity, critical infrastructure, ransomware
Management	CISO / Security Manager	Business decisions, regulatory notification, external communications	Data breach, regulatory impact, public disclosure required

3.2 Escalation Criteria

An incident should be escalated when ANY of the following conditions are met:

- Confirmed compromise of a production system or critical asset
- Multiple systems showing signs of lateral movement
- Evidence of data exfiltration (large outbound transfers, DLP alerts)
- Persistent threat actor — attacker regains access after initial containment
- Alert involves a privileged account (domain admin, root, service account)
- Incident has potential regulatory impact (PCI, HIPAA, GDPR data involved)
- Tier 1 analyst cannot determine root cause within 30 minutes

3.3 SITREP Structure (Situation Report)

A SITREP is a standardized document that communicates incident status clearly and efficiently. It must be written for TWO audiences simultaneously: technical analysts who need detail, and management who need the business impact summary.

Section	Content Required	Length	Audience
Incident Title	Clear, descriptive name + incident ID	1 line	All
Executive Summary	What happened, impact, current status (non-technical)	3-5 sentences	Management
Technical Details	Source IP, affected system, MITRE technique, timeline	1-2 paragraphs	Technical
Actions Taken	Numbered list of containment / investigation steps completed	5-10 bullets	All
Current Status	Contained / Active / Investigating / Closed	1 line	All
Next Steps	Pending tasks, owner, ETA	Numbered list	Technical
Recommendations	Long-term fixes to prevent recurrence	3-5 bullets	Management

3.4 NIST SP 800-61 Incident Response Lifecycle

NIST Special Publication 800-61 (Computer Security Incident Handling Guide) defines the standard incident response lifecycle used by most enterprise SOC teams:

Phase	Key Activities	NIST Reference	Relevant Tools
1. Preparation	Build runbooks, train staff, configure tools, establish baselines	SP 800-61 §3.1	Wazuh, Elastic, TheHive, Playbooks
2. Detection & Analysis	Identify events, triage alerts, correlate logs, determine severity	SP 800-61 §3.2	Wazuh, Elastic Security, OTX, VirusTotal

3. Containment	Short-term: isolate host. Long-term: patch, block IP, reset creds	SP 800-61 §3.3	CrowdSec, firewall rules, EDR isolation
4. Eradication	Remove malware, close attack vectors, validate clean state	SP 800-61 §3.4	Velociraptor, AV scan, reimaging
5. Recovery	Restore services, monitor for reinfection, validate normal operations	SP 800-61 §3.5	Monitoring dashboards, change management
6. Lessons Learned	Post-incident review within 2 weeks, update runbooks	SP 800-61 §3.4	Google Docs, TheHive timeline, JIRA

3.5 SOAR Automation (Splunk Phantom / XSOAR)

Security Orchestration, Automation and Response (SOAR) platforms automate repetitive SOC tasks using playbooks — pre-built workflows triggered by alert conditions. This reduces analyst fatigue and improves mean time to respond (MTTR).

Automation Use Case	Trigger	Automated Actions	Human Needed For
Alert Enrichment	New alert received	Query OTX + VT, add reputation data, update ticket	Final triage decision
IOC Blocking	High-severity malicious IP confirmed	Add to firewall blocklist via API	Approval for critical systems
Case Creation	Alert exceeds severity threshold	Create TheHive case, assign to analyst, send Slack notification	Investigation
Auto-Escalation	Case open >30 min with no action	Reassign to Tier 2, notify manager	Override if false positive
Evidence Collection	Endpoint compromise suspected	Trigger Velociraptor collection, hash evidence, preserve	Analysis of results

■ **Industry Data**	Industry benchmark: SOAR automation typically reduces Tier 1 analyst workload by 60-80% on routine alert enrichment tasks. The goal is not to replace analysts — it is to free them to focus on complex investigations that require human judgment.

3.6 Communication Templates

Manager Briefing Template (100 words):

■ **Template**	On [DATE], our security monitoring systems detected [BRIEF DESCRIPTION OF INCIDENT]. The activity originated from [SOURCE] and affected [SYSTEM/DATA]. Our security team immediately investigated and confirmed [IMPACT LEVEL]. We have [CONTAINMENT ACTIONS TAKEN] to prevent further impact. [CURRENT STATUS — e.g., 'The threat has been contained and is under investigation.']. There is currently [evidence / no evidence] of data loss. Our team is continuing to investigate and we will provide an update within [TIMEFRAME]. Recommended action: [MANAGEMENT DECISION NEEDED].

Learning Resources

- NIST SP 800-61 Rev 2: https://nvlpubs.nist.gov/nistpubs/SpecialPublications/NIST.SP.800-61r2.pdf
- SANS Incident Handler's Handbook: https://www.sans.org/reading-room/
- Splunk SOAR Documentation: https://docs.splunk.com/Documentation/SOAR
- MITRE ATT&CK; Navigator: https://mitre-attack.github.io/attack-navigator/
- TheHive Project Docs: https://docs.strangebee.com/thehive/